

Zachary Bretton

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SUMMARY

PhD in Cognitive and Computational Neuroscience with 6+ years of experience working with complex human performance and biomedical datasets. Demonstrated expertise in statistical modeling, reproducible workflow design, and scalable data analysis using Python, MATLAB, and high-performance computing. Experienced in curating and integrating multi-modal data including clinical, neuroimaging, biomarker, and high-dimensional biological measures. Proven ability to lead interdisciplinary collaborations, communicate results to technical and non-technical stakeholders, and deliver analysis and visualization outputs that support scientific discovery, publications, and grant applications.

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Ph.D. in Neuroscience

Austin, TX

2019 - 2024

Thesis: The neural mechanisms and long-term impacts of working memory suppression

BOSTON UNIVERSITY

B.A. Biology, Specialization in Neurobiology

Boston, MA

2011 - 2015

PROFESSIONAL EXPERIENCE

INSTITUTE FOR NEURODEGENERATIVE DISORDERS

New Haven, CT

Data Scientist (Clinical Research)

March 2026 - Present

- Own end-to-end curation and QC of large-scale PPMI datasets across clinical, biomarker, neuroimaging, and multi-omics modalities, ensuring integrity, traceability, and compliant downstream use.
- Develop reproducible Python pipelines to harmonize and integrate multimodal data into analysis-ready releases for internal teams and external collaborators.
- Conduct statistical and computational analyses on clinical and high-dimensional biological data, translating outputs into actionable insights for Parkinson's disease research.
- Deliver publication- and grant-quality figures, visualizations, and summaries; support collaborators with clear documentation (data dictionaries), interpretation, and Git-based reproducible workflows.

DCS CORP

Dayton, OH

Cognitive Neuroscientist (Research Scientist)

October 2024 - February 2026

- Working with the Air Force Research Labs' (AFRL) Non-Invasive Brain Stimulation team to advance neuroimaging research on cognitive performance and brain stimulation.
- Led the development of scalable data pipelines in Python and SQL for high-volume neuroimaging data, improving computational efficiency and reproducibility across projects.
- Built and validated machine learning models (classification, connectivity, pattern recognition) to extract insights from complex neuroimaging datasets and evaluate intervention outcomes.
- Reduced analysis runtime by 90% using distributed computing (Dask), enabling rapid iteration and insights.

UNIVERSITY OF TEXAS AT AUSTIN

Austin, TX

Graduate Research Assistant, Lewis-Peacock Lab

August 2019 - August 2024

- Developed and optimized predictive ML models, improving classification accuracy of high-dimensional time-series datasets through feature engineering and hyperparameter tuning, leading to multiple publications.
- Built end-to-end pipelines in Python and MATLAB for preprocessing, cleaning, and normalizing large datasets, automating repetitive tasks and improving research throughput.
- Designed and implemented rigorous experimental protocols for fMRI studies, ensuring high data integrity and reproducibility standards.
- Executed large-scale data processing on HPC clusters with Docker, SLURM, and containerization for reproducible, scalable ML workflows.

TECHNICAL SKILLS

- Proficient in Python (NumPy, SciPy, pandas, seaborn, matplotlib, nibabel, nilearn, statsmodels), MATLAB, SQL, Unix/BASH, Git, Machine Learning (scikit-learn, TensorFlow), PySpark
- Skilled in data transformation, normalization, and feature engineering; advanced statistical analysis (linear, predictive, and multivariate modeling); data visualization; high-performance computing and containerization (HPCC, Docker, Singularity, SLURM, HTCondor, Dask)